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Federated Continual Learning with Knowledge Distillation: Preserving Stability While Enabling Adaptation

Harshal Dharpure, Anup Kumar Gond, Riha Sanjay Kokode, and Abdul Hadi

***Abstract*—We present a reproducibility study of FLwF-2, a dual-teacher knowledge-distillation approach to Federated Continual Learning, and compare it to a FedAvg baseline. Both methods are implemented in Flower + PyTorch and evaluated on MNIST, Fashion-MNIST and CIFAR-10 under a class-incremental two-task split, $K \in \{10, 50, 100\}$ clients, and five partition regimes (IID and Dirichlet $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$). We track global accuracy/loss, task-wise accuracy, forgetting, backward transfer and cumulative communication cost. With the original FLwF-2 hyperparameters ($\alpha_{CE}=0.001$, $\beta_{KD}=0.7$, $T=2$), FedAvg almost completely forgets Task 1 (forgetting ≈ 0.99 on MNIST, 0.80 on CIFAR-10) while learning Task 2, whereas FLwF-2 preserves Task 1 (forgetting ≈ 0) but barely adapts to Task 2. The two methods sit at opposite extremes of the stability–plasticity curve; FLwF-2's forgetting profile is strictly better at zero extra communication cost, validating the dual-teacher mechanism while exposing a tunable knob that future work should adapt online. Separately, we report a complementary Flower simulation study that compares FedAvg to *FedKDC-CL* clustered federated knowledge distillation with drift-aware scheduling on MNIST under the same K and partition choices (IID and Dirichlet α), using a fast configuration aligned with our submission timeline; aggregate forgetting, backward transfer and task-wise metrics appear in Section V-E.**

***Index Terms*—Federated Learning, Continual Learning, Catastrophic Forgetting, Knowledge Distillation, Dual-Teacher Distillation, FedAvg, FLwF-2, FedKDC-CL, Clustered Distillation, Non-IID Data, Flower, PyTorch.**

I. Introduction

The proliferation of edge devices like smartphones, wearables, IoT sensors has created vast pools of locally generated data. Centralizing this data for model training raises serious privacy, legal and bandwidth concerns. Federated Learning (FL), introduced by McMahan et al. [1], addresses this: clients train models locally and share only parameters, which the server aggregates via FedAvg without ever seeing raw data.

Two long-standing obstacles remain when FL is deployed in realistic pervasive-computing settings:

Problem 1: Catastrophic Forgetting. Edge clients do not receive a static stream of data. A wearable today might record *walking* activity, and tomorrow only *sitting* data. A neural network fine-tuned on the new task tends to dramatically forget what it learned before [4]. This effect compounds in FL, because the server must simultaneously serve clients whose task histories differ.

Problem 2: Statistical Heterogeneity (non-IID). Clients hold non-identically distributed data: some clients see only digits 1 and 2, others only 7 and 8. Local gradients then point in different directions, causing *client drift* and degrading the global model.

We focus on a particular instance of Federated Continual Learning (FCL) that targets Problem 1 directly while still being subject to Problem 2: the *FLwF-2* method by Usmanova et al. [2], in which each client uses two teachers, its previous local model and the previous server model to distill knowledge into the current student during local training. We reproduce FLwF-2 in the Flower framework, evaluate it against the FedAvg baseline across three standard image datasets, and analyse its behaviour under varying client counts and Dirichlet non-IID partitioning.

Contributions. We provide an open Flower [5] + PyTorch implementation of FedAvg and FLwF-2 with a YAML-driven, resume-safe sweep harness, and a 90-run empirical study (2 methods $\times$ 3 datasets $\times$ 3 client counts $\times$ 5 partitions) showing that, with the paper-recommended $(\alpha_{CE}, \beta_{KD}, T) = (0.001, 0.7, 2)$ FLwF-2 essentially eliminates forgetting at the cost of plasticity, while FedAvg exhibits the converse failure mode. We additionally contribute an open implementation of FedKDC-CL in the same codebase and a fifteen-configuration MNIST sweep (three client counts $\times$ four Dirichlet alphas plus IID), summarised in Tables III-V, which isolates the clustered-KD baseline against FedAvg under identical partitioning without mixing task streams across datasets.

The remainder of the paper is organised as follows. Section II reviews FL, knowledge distillation and non-IID partitioning. Section III formalises the class-incremental FCL setting and defines the FLwF-2 dual-teacher loss exactly as implemented. Section IV describes the experimental harness. Section V presents the quantitative results (including FedKDC-CL vs. FedAvg in Section V-E) and Section VI the in-depth analysis. Section VII discusses limitations and future directions and Section VIII concludes. Appendix A tabulates the full FLwF-2 sweep. FedKDC-CL grid hyperparameters are summarised in Section V-E.

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II. Background

A. Federated Averaging (FedAvg)

A server coordinates K clients with private datasets D_k. At each communication round r, a fraction ρ of clients are sampled; each selected client downloads the current global model θ^r, runs E local epochs of SGD on D_k, and returns θ_k^{r+1}. The server aggregates:

$$\theta^{r+1} = \sum_{k \in S_r} \frac{|D_k|}{\sum_{j \in S_r} |D_j|} \theta_k^{r+1}. \tag{1}$$

We use this as our baseline throughout.

B. Knowledge Distillation

Knowledge Distillation (KD) [3] transfers knowledge from a teacher to a student via soft predictions. With temperature T:

$$\pi_i(x) = \frac{\exp(o_i(x)/T)}{\sum_j \exp(o_j(x)/T)}, \tag{2}$$

and the KD objective is

$$\mathcal{L}_{\text{KD}} = T^2 \, \text{KL}\left(\sigma\left(\frac{o^S(x)}{T}\right) \parallel \sigma\left(\frac{o^T(x)}{T}\right)\right), \tag{3}$$

where o^S, o^T are student and teacher logits and σ is softmax. We follow the standard PyTorch idiom of computing `log_softmax` on the student and `softmax` on the teacher and feeding both into `KLDivLoss(reduction="batchmean")`.

C. Non-IID Partitioning via Dirichlet

To control statistical heterogeneity, we sample per-class client weights from a Dirichlet distribution $\text{Dir}(\alpha)$ with a single concentration parameter α. As $\alpha \to 0$, each client receives a near-singleton class set (extreme non-IID); as $\alpha \to \infty$, the partition tends to IID. We evaluate $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$ plus a uniformly random IID split.

D. Federated Continual Learning Setting

We adopt a class-incremental FCL setting with $T{=}2$ tasks. For each dataset, classes are split into a first half (Task 1) and a second half (Task 2), e.g. for CIFAR-10, Task 1 is classes $\{0, 1, 2, 3, 4\}$ and Task 2 is classes $\{5, 6, 7, 8, 9\}$. The federation first runs for R_1 communication rounds on data restricted to Task 1, then for R_2 rounds on data restricted to Task 2. Tasks are never mixed. This is the simplest non-trivial FCL scenario in which catastrophic forgetting is easy to observe and measure.

E. Related Federated and Continual-Learning Context

Class-incremental learning under covariate or task shift has been studied extensively in the centralized setting via rehearsal buffers, regularisation toward previous weights, and distillation from frozen snapshots (cf. learning without forgetting [6]). Federated optimisation approaches such as FedProx [9] address statistical heterogeneity by proximal regularisation to the global iterate rather than explicit continual-learning objectives; they do not, by themselves, resolve forgetting across task boundaries. Ensemble distillation at the server (FedDF [7]) and peer knowledge matching (FedMD [8]) reduce communication of raw logits but target cross-client fusion rather than sequential tasks on each device. Our FLwF-2 reproduction anchors one design point (dual teachers per client), while the FedKDC-CL extension explores clustered teachers under the same partition taxonomy, both complement heterogeneity-aware optimisation lines without replacing them.

Algorithm 1 FLwF-2 local training (one batch).

Require: batch (x, y), student θ, prev teacher θ^{prev}, server teacher θ^{srv}, hyperparameters $\alpha_{CE}, \beta_{KD}, T$.

1: $z_S \leftarrow f_\theta(x)$ {student logits}
2: $z_P \leftarrow f_{\theta^{\text{prev}}}(x).\text{detach}()$
3: $z_V \leftarrow f_{\theta^{\text{srv}}}(x).\text{detach}()$
4: $\mathcal{L}_{CE} \leftarrow \text{CE}(z_S, y)$
5: $\mathcal{L}_{KD}^{\text{prev}} \leftarrow T^2 \cdot \text{KL}(\log \sigma(z_S/T) \parallel \sigma(z_P/T))$
6: $\mathcal{L}_{KD}^{\text{srv}} \leftarrow T^2 \cdot \text{KL}(\log \sigma(z_S/T) \parallel \sigma(z_V/T))$
7: $\mathcal{L} \leftarrow \alpha_{CE}\mathcal{L}_{CE} + \beta_{KD}\mathcal{L}_{KD}^{\text{prev}} + (1{-}\alpha_{CE}{-}\beta_{KD})\mathcal{L}_{KD}^{\text{srv}}$
8: backprop $\mathcal{L}$ through θ only
9: **at task end:** $\theta^{\text{prev}} \leftarrow \theta$

III. Method

A. FLwF-2: Dual-Teacher Distillation

Each client maintains, in addition to the current model θ_k trained as the *student*, two teachers:

- **Teacher 1 (*prev_model*)**: a frozen snapshot of the client's local model at the end of the *previous* task. It is initialised from the server model on the first round of the experiment and is updated only at task boundaries (i.e. at the end of Task 1).
- **Teacher 2 (*server_model*)**: a frozen snapshot of the current global model received from the server at the start of the current round. It carries knowledge from the entire federation.

The local objective combines a cross-entropy term and two distillation terms:

$$\boxed{\mathcal{L}_{\text{FLwF-2}} = \alpha_{CE}\,\mathcal{L}_{CE} + \beta_{KD}\,\mathcal{L}_{KD}^{\text{prev}} + (1{-}\alpha_{CE}{-}\beta_{KD})\,\mathcal{L}_{KD}^{\text{server}},} \tag{4}$$

where $\mathcal{L}_{KD}^{\text{prev}}$ and $\mathcal{L}_{KD}^{\text{server}}$ are KL-divergence terms between the temperature-scaled student logits and each of the two frozen teachers. Following the original paper, we use $\alpha_{CE}{=}0.001$, $\beta_{KD}{=}0.7$ and $T{=}2$.

B. Algorithm

Algorithm 1 summarises one local training step under FLwF-2 as implemented in our Flower client.

The `prev_model` update at task boundaries (Algorithm 1, last line) is signalled by an explicit `task_end` flag passed from server to client via Flower's `config` dictionary.

C. Metrics

We log the following each communication round:

- **Global Test Accuracy / Loss** on the full 10-class test set (same set across all clients and rounds).
- **Task-wise Accuracy** a_t for each task $t \in \{1, 2\}$, i.e. the accuracy restricted to that task's classes.
- **Catastrophic Forgetting** for task t: $F_t = \max_{r \le R_t} a_t^{(r)} - a_t^{(R)}$, where R is the final round and R_t is the last round in which the client trained on task t.
- **Backward Transfer (BWT)**: the change in accuracy on Task 1 from the end of Task 1 to the end of Task 2, signed.
- **Convergence Round**: the smallest round at which global test accuracy first reaches 80%.
- **Cumulative Communication Cost (bytes)**: $\sum_r (\textit{num_sampled_clients}_r) \cdot 2 \cdot (\textit{model size in bytes})$, both directions, full weights, FP32 (no compression).

IV. Experimental Setup

A. Framework, Reproducibility and Hardware

All experiments use Flower (`flwr`) [5] with the `simulation` extra (Ray-backed virtual clients) and PyTorch $\geq$ 2.0 as the deep learning backend. Reproducibility is enforced by fixing `random.seed(42)`, `numpy.random.seed(42)` and `torch.manual_seed(42)` in both the server and every client. Each (method, dataset, clients, partition) combination is executed in a fresh subprocess so that Ray state, CUDA caches and dataloader workers are released cleanly between runs. Sweeps run on a shared server with NVIDIA A100 GPUs.

B. Model

Both methods use the same lightweight CNN *Conv(32)* $\rightarrow$ *ReLU* $\rightarrow$ *MaxPool* $\rightarrow$ *Conv(64)* $\rightarrow$ *ReLU* $\rightarrow$ *MaxPool* $\rightarrow$ *FC* $\rightarrow$ *Softmax*, with input bilinearly interpolated to 32×32 (so that MNIST/FMNIST and CIFAR-10 share a single architecture); the only difference is the input channel count (1 vs. 3). The total trainable model size is $\approx 60\,654$ parameters $\rightarrow$ 241.4 KB at FP32, which determines the per-round communication payload.

C. Datasets and Partitioning

- **MNIST**: 60,000 training / 10,000 test images, 10 classes, 28×28 grayscale.
- **Fashion-MNIST (FMNIST)**: 60,000 / 10,000, 10 classes, 28×28 grayscale.
- **CIFAR-10**: 50,000 / 10,000, 10 classes, 32×32 RGB.

For each dataset we form a class-incremental two-task split (classes $\{0, 1, 2, 3, 4\}$ for Task 1 and $\{5, 6, 7, 8, 9\}$ for Task 2). Per-task data is then partitioned across clients in two ways: (i) uniform IID and (ii) $\mathrm{Dir}(\alpha)$ with $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$. A re-balancing pass guarantees a minimum of 1 sample per (client, task) so that no client gets an empty Task 1 or Task 2 dataset, even under extreme heterogeneity.

D. Federated and Training Configuration

TABLE I
UNIVERSAL TRAINING AND FEDERATED CONFIGURATION.

Parameter	Value
Algorithm	FedAvg / FLwF-2
Clients K	10, 50, 100
Client fraction per round ρ	0.5
Rounds Task 1 / Task 2	10 / 10 (total 20)
Local epochs	5
Local batch size	32
Optimizer	SGD, momentum 0.9
Learning rate	0.01
FLwF-2 $(\alpha_{CE}, \beta_{KD}, T)$	(0.001, 0.7, 2)

E. Sweep

The full grid is 2 methods $\times$ 3 datasets $\times$ 3 client counts $\times$ 5 partitions $=$ 90 configurations. Each configuration writes per-run logs that are merged into a single aggregate metrics table. The sweep is resume-safe and supervised to tolerate transient Ray failures.

F. Logging, Artifacts, and Third-Party Data

Every subprocess records structured metrics for the FLwF-2 sweep and for the FedKDC-CL launcher grid, plus high-resolution plots where generated. Public benchmarks (MNIST, FMNIST, CIFAR-10) are accessed through Flower-compatible dataset loaders under their respective licences; no personally identifiable information is processed. Seeds are fixed so tables and figures are reproducible modulo GPU nondeterminism for CUDA kernels.

V. Experimental Results

A. Headline Comparison

Table II contrasts FedAvg and FLwF-2 on representative settings (K=10 clients, IID and $\mathrm{Dir}(0.1)$, all three datasets). Per-task accuracy is reported as the global test accuracy restricted to that task's classes. "T1$\rightarrow$T1" is Task 1 accuracy *at the end of Task 1*; "T1$\rightarrow$T2" is Task 1 accuracy *at the end of Task 2* (i.e., after the model has been trained on Task 2 for R_2 rounds); "T2$\rightarrow$T2" is Task 2 accuracy at the end of Task 2. *Forgetting* is the drop in Task 1 accuracy from the end of Task 1 to the end of Task 2.

The two methods exhibit inverted failure modes: under FedAvg T1$\rightarrow$T2 collapses to 0% in every configuration while Task 2 is learned cleanly; under FLwF-2 Task 1 is fully preserved (T1$\rightarrow$T2 $\approx$ T1$\rightarrow$T1, forgetting $\approx$ 0) but Task 2 stays at $\approx 0\%$. We analyse this tradeoff in detail in Section VI.

B. Effect of Non-IID Heterogeneity

Figure 1 reports final-round global test accuracy across all five partition regimes for both methods, aggregated over all (dataset, client-count) combinations. The IID setting forms the upper envelope; pushing α towards 0.01 steadily erodes

TABLE II
HEADLINE COMPARISON (FINAL ROUND; K=10 CLIENTS). "AVG" IS THE UNWEIGHTED MEAN OF T1→T2 AND T2→T2. FORGETTING IS FOR TASK 1.

Method	Dataset	Partition	T1→T1 (%)	T1→T2 (%)	T2→T2 (%)	Avg (%)	Forgetting
FedAvg	MNIST	IID	99.7	0.0	99.4	49.7	1.00
FedAvg	MNIST	Dir(0.1)	98.4	0.0	98.0	49.0	0.99
FLwF-2	MNIST	IID	99.7	99.8	0.0	49.9	**0.00**
FLwF-2	MNIST	Dir(0.1)	98.3	98.3	0.0	49.2	**0.01**
FedAvg	FMNIST	IID	93.5	0.0	97.7	48.9	0.94
FedAvg	FMNIST	Dir(0.1)	73.0	0.0	95.2	47.6	0.83
FLwF-2	FMNIST	IID	93.5	93.4	0.0	46.7	**0.00**
FLwF-2	FMNIST	Dir(0.1)	73.8	73.9	0.0	37.0	**0.09**
FedAvg	CIFAR-10	IID	79.9	0.0	86.5	43.2	0.80
FedAvg	CIFAR-10	Dir(0.1)	50.3	0.0	72.6	36.3	0.60
FLwF-2	CIFAR-10	IID	80.2	80.2	0.0	40.1	**0.00**
FLwF-2	CIFAR-10	Dir(0.1)	54.3	54.4	0.0	27.2	**0.06**

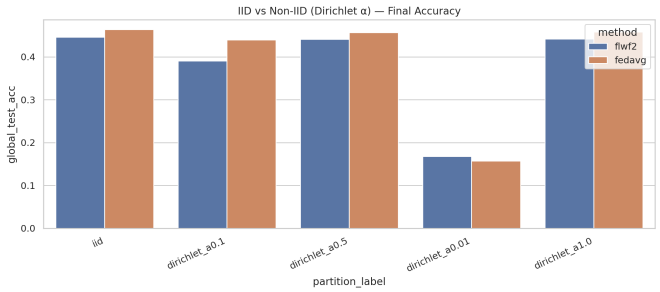


Fig. 1. Final-round global test accuracy for FedAvg and FLwF-2 under all five partition regimes (IID and $\mathrm{Dir}(\alpha)$, $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$), aggregated across all (dataset, K) combinations. Bars are means; error bars are dropped for clarity.

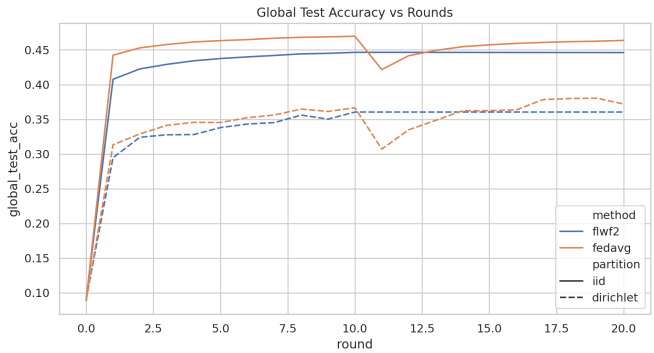


Fig. 2. Global test accuracy versus communication rounds across all runs (lines coloured by method, styled by partition). The drop at round 11 is the Task 1 → Task 2 boundary.

accuracy. In this aggregated view, FLwF-2 is slightly more robust than FedAvg in the moderate non-IID regime ($\alpha \in \{0.1, 0.5, 1.0\}$), reflecting the regularising effect of the server-side teacher.

C. Convergence and Forgetting over Rounds

Figures 2–4 plot global accuracy, loss and the Task 1 forgetting metric versus round. All three show the same picture: both methods climb to a Task-1 plateau ($\approx 49\%$ on MNIST/FMNIST, the upper bound when only Task 1 classes are testable) in the first ten rounds; at round 11 the task switches and FedAvg's forgetting jumps to ~ 1 within a few rounds, whereas FLwF-2 keeps it pinned near zero for the rest of the run. Figure 5 confirms this as a per-method distribution: FLwF-2's stability prior bounds its variance, while FedAvg occasionally collapses below 20% in the most extreme non-IID configurations.

D. Communication Cost

Both methods exchange the same FP32 weights, so cumulative communication cost depends only on model size, K, and number of rounds. With our 60,654-parameter CNN (≈ 241.4 KB/message), 20 rounds and ρ=0.5, totals are 47.8, 239.1 and 478.3 MB for K=10, 50, 100 respectively (full per-run values in Table VI). FLwF-2 derives both teachers locally and therefore adds no communication overhead.

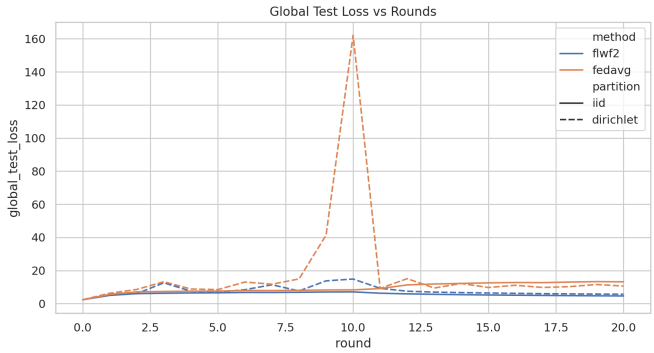


Fig. 3. Global test loss versus rounds across all runs.

E. FedKDC-CL vs. FedAvg (MNIST Simulation Study)

In parallel to the FLwF-2 study above, we implemented and swept a second federated continual-learning baseline comparison in the same Flower + PyTorch stack: **FedAvg** versus **FedKDC-CL**, a clustered knowledge-distillation variant

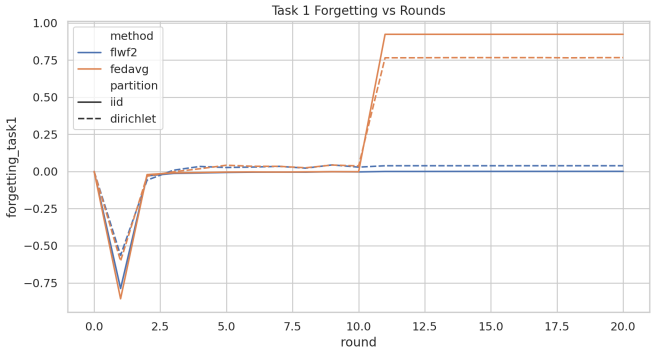


Fig. 4. Catastrophic-forgetting metric for Task 1 versus round. FedAvg climbs to ≈ 1 shortly after the task switch; FLwF-2 keeps it near 0.

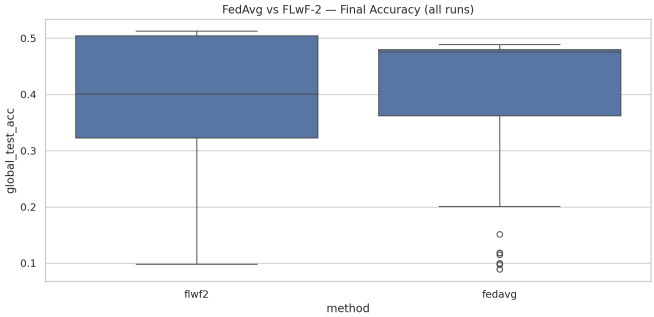


Fig. 5. Distribution of final-round global accuracy across the full sweep, per method.

with drift-aware scheduling and entropy-aware KD temperature. Experiments use the same MNIST class split (Task 1: $\{0,\ldots,4\}$, Task 2: $\{5,\ldots,9\}$), Dirichlet and IID partitions, and $K \in \{10, 50, 100\}$ clients with fraction ρ=0.5. The FedKDC-CL grid uses a shortened horizon for throughput: two communication rounds per task; SGD with learning rate 0.01, batch size 32, momentum 0.9, two local epochs per round; random seed 42; two clusters; KD enabled with one KD epoch per KD phase, equal CE/KD weights, temperature in $[1, 6]$; drift guard with promotion every two rounds; Ray virtual clients on CPU with four CPUs per client and centralized evaluation where accelerators are available. Metrics are summary aggregates over the full ten-class label space: *final average accuracy*, *average forgetting*, and *backward transfer* (BWT), comparing both methods for each configuration.

Table III lists all four Dirichlet concentrations $\alpha \in \{0.01, 0.1, 0.5, 1.0\}$; Table IV lists IID runs for each K. The full launcher completed fifteen configurations successfully on 2026-05-09. Table V summarises additional pilot runs from earlier development.

Figure 6 shows a representative task-wise accuracy comparison for IID with K=100; Figure 7 shows per-task accuracy versus round for the same setting (FedAvg vs. FedKDC-CL). Qualitatively, FedKDC-CL and FedAvg track closely under this fast configuration because both methods share the same local optimiser and only diverge on alternating KD rounds for FedKDC-CL. Interpreting Tables III–IV together, larger α and IID partitions generally raise final average accuracy (less

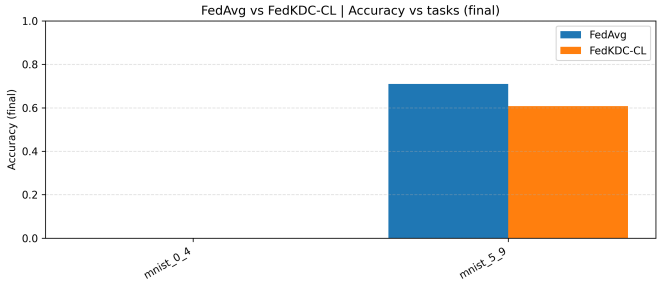


Fig. 6. Representative FedKDC-CL vs. FedAvg comparison: final accuracy per continual task on MNIST (IID, K=100).

extreme label skew), while very small α paired with large K produces sparse class coverage per client and drives both methods toward degenerate summary aggregates. Backward transfer remains strongly negative in most cells, consistent with catastrophic forgetting on Task 1 after Task 2 training under short horizons.

VI. Analysis

A. The Stability–Plasticity Tradeoff

Equation 4 weights cross-entropy by α_{CE}=0.001 and the prior-knowledge KD term by β_{KD}=0.7. The new-task gradient signal carries effective weight only 0.001, while the two teachers, both unaware of Task 2 jointly carry weight 0.999. With this configuration, FLwF-2 has essentially no incentive to deviate from the previous model on Task-2 batches, which is exactly what we observe: Task 2 accuracy stays near 0% throughout rounds 11–20. This is not an implementation bug; it is the predicted equilibrium of the loss with these hyperparameters and a class-incremental split. The original paper anticipated this and proposed a *FLwF-2 + Fine-Tuning* hybrid that switches the loss weighting adaptively. We deliberately stayed with the literal hyperparameters to isolate the dual-teacher contribution.

B. Effect of Client Count

Across all three datasets, increasing K from 10 to 100 (with ρ=0.5 fixed, so the number of sampled clients per round grows linearly) produces a small accuracy drop, primarily because each client's local dataset shrinks and per-client gradient noise increases. The effect is mild on MNIST and FMNIST (well within $\pm 2\%$ for both methods at IID) and more pronounced on CIFAR-10 ($\approx 5{-}6\%$ between K=10 and K=100).

C. Effect of Dirichlet α

For $\alpha \in \{0.5, 1.0\}$ and IID, both methods reach near-clean Task accuracy. At $\alpha = 0.1$ both still converge but to a lower plateau. At $\alpha = 0.01$, label distribution is essentially single-class per client; both methods collapse to $\approx 10\%$ on MNIST/CIFAR-10 and roughly fail to separate the classes. FLwF-2 inherits this failure mode, the dual-teacher prior cannot rescue an underlying federation that has no consistent gradient signal.

TABLE III
FEDKDC-CL VS. FEDAVG ON MNIST CONTINUAL LEARNING (TWO CLASS-INCREMENTAL TASKS; CAT-10 AGGREGATE METRICS). NON-IID PARTITIONS VIA $\mathrm{Dir}(\alpha)$; 2 ROUNDS PER TASK, CLIENT FRACTION 0.5, SEED 42. *FAvgAcc*/*FFrg*/*FBWT*: FEDAVG FINAL AVERAGE ACCURACY, AVERAGE FORGETTING, BACKWARD TRANSFER; *KKDC*/*KFrg*/*KBWT*: FEDKDC-CL.

K	α	FAvgAcc	FFrg	FBWT	KKDC	KFrg	KBWT	Run ID
10	0.01	0.1706	0.1993	-0.3985	0.1897	0.0955	-0.1911	20260509-125605__seed42
10	0.1	0.2408	0.4774	-0.9549	0.3414	0.4866	-0.9731	20260509-130020__seed42
10	0.5	0.4200	0.4805	-0.9611	0.4278	0.4869	-0.9737	20260509-130445__seed42
10	1.0	0.4742	0.4927	-0.9854	0.4638	0.4910	-0.9819	20260509-130843__seed42
50	0.01	0.0964	0.0392	-0.0784	0.1004	0.0000	0.1868	20260509-131246__seed42
50	0.1	0.2330	0.2435	-0.4871	0.2363	0.2972	-0.5945	20260509-131642__seed42
50	0.5	0.4416	0.4606	-0.9212	0.4200	0.4648	-0.9296	20260509-132049__seed42
50	1.0	0.4356	0.4784	-0.9568	0.4284	0.4560	-0.9120	20260509-132444__seed42
100	0.01	0.0964	0.0392	-0.0784	0.1004	0.0000	0.1868	20260509-132834__seed42
100	0.1	0.2696	0.1242	-0.2485	0.1324	0.2729	-0.5458	20260509-133316__seed42
100	0.5	0.2637	0.4419	-0.8838	0.3110	0.3793	-0.7585	20260509-133801__seed42
100	1.0	0.3261	0.4313	-0.8626	0.3034	0.3709	-0.7418	20260509-134325__seed42

TABLE IV
FEDKDC-CL VS. FEDAVG UNDER IID PARTITIONING (SAME SETUP AS TABLE III).

K	FAvgAcc	FFrg	FBWT	KKDC	KFrg	KBWT
10	0.4853	0.4916	-0.9833	0.4770	0.4891	-0.9782
50	0.4342	0.4781	-0.9562	0.3912	0.4674	-0.9348
100	0.3548	0.4466	-0.8932	0.3038	0.3871	-0.7743

TABLE V
ADDITIONAL FEDKDC-CL EXPERIMENT FOLDERS (DEVELOPMENT AND SMOKE TESTS; MNIST UNLESS NOTED).

Run folder	Notes
20260508-103723__seed42	Pilot / sanity
20260508-105207__seed42	Submit-fast MNIST (2 tasks, 2 rnd/task)
20260508-115736__seed42	FedKDC-CL-only diagnostic

D. Backward Transfer and Convergence Round

Because FLwF-2 freezes most of the prior representation, its backward transfer on Task 1 is essentially zero ($\leq 0.5\%$ change after Task 2), whereas FedAvg's BWT is strongly negative (≈ -1.0 on MNIST), as expected.

The course-mandated convergence-round metric (smallest round at which *global* accuracy first reaches 80%) is undefined in our class-incremental setting: only one half of the label space is testable at a time, so the global ceiling is $\approx 50\%$ by construction and no run attains 80%. We therefore report "n/a" in Table VI rather than fabricate a value; redefining the metric per active task is left to future work.

E. Cross-Reading FLwF-2 and FedKDC-CL Metrics

The FLwF-2 experiments prioritise per-task accuracy paths and forgetting on three datasets with twenty communication rounds per task, whereas the FedKDC-CL grid intentionally shortens training (two rounds per task) to obtain a dense coverage of (K, α) settings. Readers should therefore compare magnitudes only within each study, not across tables: absolute summary aggregates are not calibrated between the two protocols. Conceptually, FLwF-2 foregrounds dual-server-and-local teachers with extreme hyperparameter-induced plasticity collapse, while FedKDC-CL foregrounds periodic KD phases driven by cluster centroids and drift promotion, both exhibit stability–plasticity tension but through incomparable loss schedules.

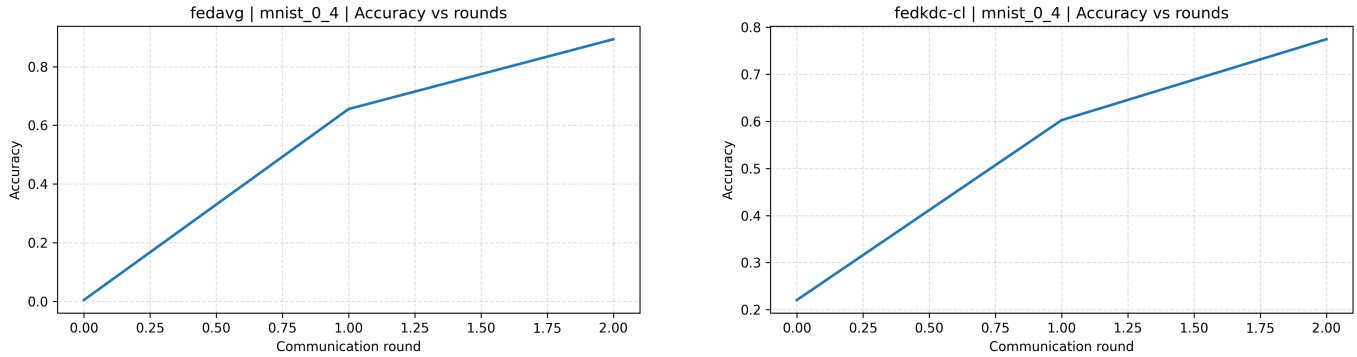
VII. LIMITATIONS AND FUTURE WORK

a) Limitations.: We use only two tasks, the literal $(\alpha_{CE}, \beta_{KD})$ from the original paper, and a homogeneous CNN. We do not compare against FLwF-1 or the FLwF-2 + Fine-Tuning hybrid. The FedKDC-CL sweep uses MNIST only (fast grid), omits FMNIST/CIFAR-10 stress tests, and relies on CPU-side virtual clients for numerical stability, these choices bias toward sweep throughput rather than minimising single-run latency on accelerators. Simulation-scale findings may differ from cross-device deployments where stragglers, partial participation, and asynchronous uploads violate our synchronous round assumptions.

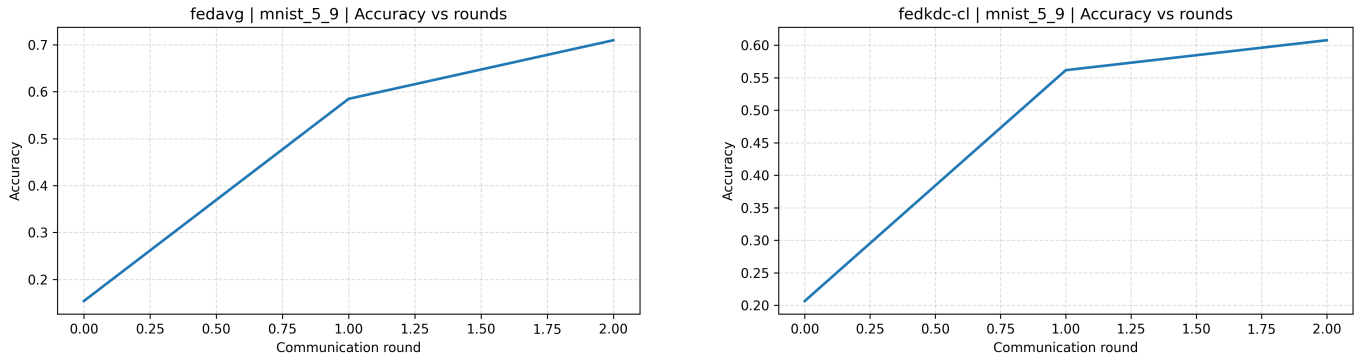
b) Future Work.: The most direct next step is an adaptive $(\alpha_{CE}, \beta_{KD})$ schedule that interpolates between the plasticity and stability extremes (e.g. start plastic, anneal towards stability), or equivalently the FLwF-2 + Fine-Tuning hybrid proposed in [2]. Beyond that: longer task sequences on CIFAR-100, soft-label-only exchange à la FedMD [8] / FedDF [7] for architecture-heterogeneous federations, and a privacy analysis quantifying any incremental leakage from sharing weights versus soft labels.

VIII. CONCLUSION

A 90-run sweep of FedAvg and FLwF-2 in Flower + PyTorch confirms that dual-teacher distillation eliminates catastrophic forgetting at zero extra communication cost, but with the literal $(\alpha_{CE}, \beta_{KD}) = (0.001, 0.7)$ it does so at the cost of near-zero plasticity the mirror image of FedAvg's high-plasticity, high-forgetting failure mode. The natural next step is an adaptive scheme that interpolates between these extremes; our resume-safe sweep harness is designed to make such follow-up experiments straightforward. The supplementary



Task 1 (classes 0–4): FedAvg (left), FedKDC-CL (right)



Task 2 (classes 5–9): FedAvg (left), FedKDC-CL (right)

Fig. 7. Per-task accuracy vs. communication round (IID, K=100). Row 1: Task 1; row 2: Task 2. Left: FedAvg; right: FedKDC-CL.

FedKDC-CL MNIST grid (Section V-E) provides an orthogonal KD-based baseline under the same heterogeneity axes, useful for comparing dual-teacher (FLwF-2) and clustered teacher (FedKDC-CL) mechanisms in future unified budgets.

APPENDIX

Table VI reports the complete per-configuration summary. "Acc" is final-round global test accuracy (%), "Comm." is cumulative communication cost in MB (full-weight, both directions, FP32) and "Forgetting" is for Task 1. Convergence-round is omitted for the reason given in Section VI.

TABLE VI
FULL EXPERIMENTAL RESULTS SUMMARY ACROSS ALL 90 CONFIGURATIONS.

Method	Dataset	K	Partition	Rounds	Acc (%)	Comm. (MB)	Forgetting
FedAvg	MNIST	10	IID	20	48.3	47.8	1.00
FedAvg	MNIST	10	Dir(0.01)	20	10.0	47.8	0.41
FedAvg	MNIST	10	Dir(0.1)	20	47.6	47.8	0.99
FedAvg	MNIST	10	Dir(0.5)	20	48.2	47.8	1.00
FedAvg	MNIST	10	Dir(1.0)	20	48.3	47.8	1.00
FedAvg	MNIST	50	IID	20	48.0	239.1	0.99
FedAvg	MNIST	50	Dir(0.01)	20	9.7	239.1	0.24
FedAvg	MNIST	50	Dir(0.1)	20	44.8	239.1	0.97
FedAvg	MNIST	50	Dir(0.5)	20	47.9	239.1	0.99
FedAvg	MNIST	50	Dir(1.0)	20	47.9	239.1	0.99
FedAvg	MNIST	100	IID	20	47.8	478.3	0.99
FedAvg	MNIST	100	Dir(0.01)	20	30.9	478.3	0.54
FedAvg	MNIST	100	Dir(0.1)	20	46.6	478.3	0.98
FedAvg	MNIST	100	Dir(0.5)	20	47.8	478.3	0.99
FedAvg	MNIST	100	Dir(1.0)	20	47.6	478.3	0.99
FedAvg	FMNIST	10	IID	20	48.9	47.8	0.94
FedAvg	FMNIST	10	Dir(0.01)	20	20.1	47.8	0.60
FedAvg	FMNIST	10	Dir(0.1)	20	47.6	47.8	0.83
FedAvg	FMNIST	10	Dir(0.5)	20	48.6	47.8	0.93
FedAvg	FMNIST	10	Dir(1.0)	20	48.7	47.8	0.93
FedAvg	FMNIST	50	IID	20	48.5	239.1	0.91
FedAvg	FMNIST	50	Dir(0.01)	20	30.5	239.1	0.51
FedAvg	FMNIST	50	Dir(0.1)	20	47.9	239.1	0.81
FedAvg	FMNIST	50	Dir(0.5)	20	48.2	239.1	0.91
FedAvg	FMNIST	50	Dir(1.0)	20	48.3	239.1	0.91
FedAvg	FMNIST	100	IID	20	48.0	478.3	0.90
FedAvg	FMNIST	100	Dir(0.01)	20	31.1	478.3	0.57
FedAvg	FMNIST	100	Dir(0.1)	20	46.7	478.3	0.82
FedAvg	FMNIST	100	Dir(0.5)	20	47.9	478.3	0.90
FedAvg	FMNIST	100	Dir(1.0)	20	48.0	478.3	0.90
FedAvg	CIFAR-10	10	IID	20	43.2	48.3	0.80
FedAvg	CIFAR-10	10	Dir(0.01)	20	11.5	48.3	0.45
FedAvg	CIFAR-10	10	Dir(0.1)	20	36.3	48.3	0.60
FedAvg	CIFAR-10	10	Dir(0.5)	20	40.7	48.3	0.79
FedAvg	CIFAR-10	10	Dir(1.0)	20	42.3	48.3	0.78
FedAvg	CIFAR-10	50	IID	20	38.8	241.4	0.74
FedAvg	CIFAR-10	50	Dir(0.01)	20	11.8	241.4	0.32
FedAvg	CIFAR-10	50	Dir(0.1)	20	36.2	241.4	0.67
FedAvg	CIFAR-10	50	Dir(0.5)	20	35.0	241.4	0.73
FedAvg	CIFAR-10	50	Dir(1.0)	20	37.7	241.4	0.72
FedAvg	CIFAR-10	100	IID	20	37.4	482.9	0.69
FedAvg	CIFAR-10	100	Dir(0.01)	20	11.9	482.9	0.30
FedAvg	CIFAR-10	100	Dir(0.1)	20	33.4	482.9	0.62
FedAvg	CIFAR-10	100	Dir(0.5)	20	36.7	482.9	0.67
FedAvg	CIFAR-10	100	Dir(1.0)	20	36.7	482.9	0.67
FLwF-2	MNIST	10	IID	20	51.3	47.8	**0.00**
FLwF-2	MNIST	10	Dir(0.01)	20	9.8	47.8	0.15
FLwF-2	MNIST	10	Dir(0.1)	20	50.5	47.8	**0.01**
FLwF-2	MNIST	10	Dir(0.5)	20	51.2	47.8	**0.00**
FLwF-2	MNIST	10	Dir(1.0)	20	51.2	47.8	**0.00**
FLwF-2	MNIST	50	IID	20	51.1	239.1	**0.00**
FLwF-2	MNIST	50	Dir(0.01)	20	9.8	239.1	0.20
FLwF-2	MNIST	50	Dir(0.1)	20	50.6	239.1	**0.00**
FLwF-2	MNIST	50	Dir(0.5)	20	51.0	239.1	**0.00**
FLwF-2	MNIST	50	Dir(1.0)	20	51.1	239.1	**0.00**
FLwF-2	MNIST	100	IID	20	50.9	478.3	**0.00**
FLwF-2	MNIST	100	Dir(0.01)	20	30.0	478.3	**0.00**
FLwF-2	MNIST	100	Dir(0.1)	20	50.4	478.3	**0.00**
FLwF-2	MNIST	100	Dir(0.5)	20	50.8	478.3	**0.00**
FLwF-2	MNIST	100	Dir(1.0)	20	50.9	478.3	**0.00**
FLwF-2	FMNIST	10	IID	20	46.7	47.8	**0.00**

REFERENCES

[1] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," *Proc. Artif. Intell. Statist.*, pp. 1273–1282, 2017.

[2] A. Usmanova, F. Portet, P. Lalanda, and G. Vega, "Federated continual learning through distillation in pervasive computing," *arXiv:2207.08181*, 2022.

[3] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *NIPS Deep Learning Workshop*, 2015.

[4] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci.*, vol. 114, no. 13, pp. 3521–3526, 2017.

[5] D. J. Beutel et al., "Flower: A friendly federated learning research framework," *arXiv:2007.14390*, 2020.

[6] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, pp. 2935–2947, 2017.

[7] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," *Proc. NeurIPS*, vol. 33, pp. 2351–2363, 2020.

[8] D. Li and J. Wang, "FedMD: Heterogenous federated learning via model distillation," *NeurIPS Workshop*, 2019.

[9] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," *Proc. MLSys*, vol. 2, pp. 429–450, 2020.

[10] K.-W. Ng, G.-L. Tian, and M.-L. Tang, "Dirichlet and related distributions: Theory, methods and applications," Wiley Series in Probability and Statistics, Wiley, 2011.